```
┌──(juhee㉿kali)-[~]
└─$ pwd
/home/juhee

┌──(juhee㉿kali)-[~]
└─$ ls
copy.txt   juhee.txt            myfile.txt      Pictures  report      Videos
Desktop    linux_admin_practice2  myfile.txt.save  project   student.txt
Documents  linuxlab               myfolder        Projects  Templates
Downloads  Music                  myprogram.txt   Public    test

┌──(juhee㉿kali)-[~]
└─$ ls -a ~
.               .face               myfile.txt.save          Templates
..              .face.icon          myfolder                 test
.bash_logout    .file.py.swp        myprogram.txt            Videos
.bashrc         .gnupg              Pictures                 .viminfo
.bashrc.original  .ICEauthority     .profile                 .Xauthority
.cache          .java               project                  .xsession-errors
.config         juhee.txt           Projects                 .xsession-errors.old
copy.txt        linux_admin_practice2  Public                .zprofile
Desktop         linuxlab            report                   .zsh_history
.dmrc           .local              .ssh                     .zshrc
Documents       Music               student.txt
Downloads       myfile.txt          .sudo_as_admin_successful

┌──(juhee㉿kali)-[~]
```

```
└─$ ls -lh
total 72K
-rw-rw-r-- 1 juhee juhee    0 Aug 24 05:34 copy.txt
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Desktop
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Documents
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Downloads
-rw-rw-r-- 1 juhee juhee    0 Aug 24 05:30 juhee.txt
drwxrwxr-x 2 juhee juhee 4.0K Aug 31 05:00 linux_admin_practice2
drwxrwxr-x 6 juhee juhee 4.0K Oct  5 05:26 linuxlab
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Music
-rw-rw-r-- 1 juhee juhee    0 Aug 24 05:16 myfile.txt
-rw-rw-r-- 1 juhee juhee   55 Aug 24 05:14 myfile.txt.save
drwxrwxr-x 2 juhee juhee 4.0K Aug 24 05:29 myfolder
-rw-rw-r-- 1 juhee juhee   21 Aug 24 05:19 myprogram.txt
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Pictures
drwxrwxr-x 2 juhee juhee 4.0K Aug 18 03:19 project
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Projects
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Public
drwxrwxr-x 2 juhee juhee 4.0K Aug 24 05:07 report
-rw-rw-r-- 1 juhee juhee  145 Sep 11 01:20 student.txt
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Templates
drwxrwxr-x 2 juhee juhee 4.0K Sep 18 01:06 test
drwxr-xr-x 2 juhee juhee 4.0K Aug 17 07:18 Videos

┌──(juhee㉿kali)-[~]
└─$ mkdir myfristlab
```

```
┌──(juhee㉿kali)-[~]
└─$ cd myfristlab

┌──(juhee㉿kali)-[~/myfristlab]
└─$ touch a.txt b.txt c.txt

┌──(juhee㉿kali)-[~/myfristlab]
└─$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.2"
VERSION="2026.2"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

┌──(juhee㉿kali)-[~/myfristlab]
└─$ whoami
juhee

┌──(juhee㉿kali)-[~/myfristlab]
└─$ hostname
```

kali

```
┌──(juhee㉿kali)-[~/myfristlab]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            951M     0  951M   0% /dev
tmpfs           197M  1.2M  196M   1% /run
/dev/sda1        19G   15G  3.0G  83% /
tmpfs           985M  4.0K  985M   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           985M   52K  985M   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M  108K  197M   1% /run/user/1000

┌──(juhee㉿kali)-[~/myfristlab]
└─$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       848Mi       547Mi        15Mi       739Mi       1.1Gi
Swap:          1.1Gi          0B       1.1Gi

┌──(juhee㉿kali)-[~/myfristlab]
└─$ uptime
 05:02:49 up 16 min,  1 user,  load average: 0.20, 0.24, 0.40

┌──(juhee㉿kali)-[~/myfristlab]
└─$ uname -a
```

Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux

┌──(juhee㉿kali)-[~/myfristlab]

└─$ man ls

┌──(juhee㉿kali)-[~/myfristlab]

└─$ cp --help

Usage: cp [OPTION]... [-T] SOURCE DEST

  or: cp [OPTION]... SOURCE... DIRECTORY

  or: cp [OPTION]... -t DIRECTORY SOURCE...

Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.

Mandatory arguments to long options are mandatory for short options too.

  -a, --archive

        same as -dR --preserve=all

      --attributes-only

        don't copy the file data, just the attributes

      --backup[=CONTROL]

        make a backup of each existing destination file

  -b

        like --backup but does not accept an argument

      --copy-contents

        copy contents of special files when recursive

  -d

        same as --no-dereference --preserve=links

      --debug

        explain how a file is copied. Implies -v

-f, --force

if an existing destination file cannot be opened, remove it and try again (this option is ignored when the -n option is also used)

-i, --interactive

prompt before overwrite (overrides a previous -n option)

-H

follow command-line symbolic links in SOURCE

-L, --dereference

always follow symbolic links in SOURCE

-P, --no-dereference

never follow symbolic links in SOURCE

--keep-directory-symlink

follow existing symlinks to directories

-l, --link

hard link files instead of copying

-n, --no-clobber

(deprecated) silently skip existing files. See also --update

-p

same as --preserve=mode,ownership,timestamps

--preserve[=ATTR_LIST]

preserve the specified attributes

--no-preserve=ATTR_LIST

don't preserve the specified attributes

--parents

use full source file name under DIRECTORY

-R, -r, --recursive

copy directories recursively

--reflink[=WHEN]

control clone/CoW copies. See below

--remove-destination

remove each existing destination file before attempting to open it

(contrast with --force)

--sparse=WHEN

control creation of sparse files. See below

--strip-trailing-slashes

remove any trailing slashes from each SOURCE argument

-s, --symbolic-link

make symbolic links instead of copying

-S, --suffix=SUFFIX

override the usual backup suffix

-t, --target-directory=DIRECTORY

copy all SOURCE arguments into DIRECTORY

-T, --no-target-directory

treat DEST as a normal file

--update[=UPDATE]

control which existing files are updated;

UPDATE={all,none,none-fail,older(default)}

-u

equivalent to --update[=older]. See below

-v, --verbose

explain what is being done

-x, --one-file-system

stay on this file system

-Z

set SELinux security context of destination file to default type

--context[=CTX]

like -Z, or if CTX is specified then set the
SELinux or SMACK security context to CTX

--help
display this help and exit

--version
output version information and exit

ATTR_LIST is a comma-separated list of attributes. Attributes are 'mode' for permissions (including any ACL and xattr permissions), 'ownership' for user and group, 'timestamps' for file timestamps, 'links' for hard links, 'context' for security context, 'xattr' for extended attributes, and 'all' for all attributes.

By default, sparse SOURCE files are detected by a crude heuristic and the corresponding DEST file is made sparse as well. That is the behavior selected by --sparse=auto. Specify --sparse=always to create a sparse DEST file whenever the SOURCE file contains a long enough sequence of zero bytes. Use --sparse=never to inhibit creation of sparse files.

UPDATE controls which existing files in the destination are replaced.
'all' is the default operation when an --update option is not specified,
and results in all existing files in the destination being replaced.
'none' is like the --no-clobber option, in that no files in the
destination are replaced, and skipped files do not induce a failure.
'none-fail' also ensures no files are replaced in the destination,
but any skipped files are diagnosed and induce a failure.
'older' is the default operation when --update is specified, and results
in files being replaced if they're older than the corresponding source file.

By default or with --reflink=auto, cp will try a lightweight copy, where the
data blocks are copied only when modified, falling back to a standard copy
if this is not possible. With --reflink[=always] cp will fail if CoW is not
supported, while --reflink=never ensures a standard copy is performed.

The backup suffix is '~', unless set with --suffix or SIMPLE_BACKUP_SUFFIX.
The version control method may be selected via the --backup option or through
the VERSION_CONTROL environment variable. Here are the values:

  none, off   never make backups (even if --backup is given)
  numbered, t   make numbered backups
  existing, nil   numbered if numbered backups exist, simple otherwise
  simple, never   always make simple backups

As a special case, cp makes a backup of SOURCE when the force and backup
options are given and SOURCE and DEST are the same name for an existing,
regular file.

Report bugs to: bug-coreutils@gnu.org
GNU coreutils home page: <https://www.gnu.org/software/coreutils/>
General help using GNU software: <https://www.gnu.org/gethelp/>
Full documentation <https://www.gnu.org/software/coreutils/cp>
or available locally via: info '(coreutils) cp invocation'

┌──(juhee㉿kali)-[~/myfristlab]
└─$ whatis mv
mv (1)   - move (rename) files

```
┌──(juhee㉿kali)-[~/myfristlab]
└─$ cd ~

┌──(juhee㉿kali)-[~]
└─$ mkdir -p project/src/data

┌──(juhee㉿kali)-[~]
└─$ rm myfristlab/c.txt

┌──(juhee㉿kali)-[~]
└─$ echo " Rahul"> students.txt

┌──(juhee㉿kali)-[~]
└─$ echo " Priya ">> students.txt

┌──(juhee㉿kali)-[~]
└─$ echo " Aman">>students.txt

┌──(juhee㉿kali)-[~]
└─$ echo " Riya">>students.txt

┌──(juhee㉿kali)-[~]
└─$ echo " Rohit">>students.txt
```

```
┌──(juhee㉿kali)-[~]
└─$ head -n 2 studnts.txt
head: cannot open 'studnts.txt' for reading: No such file or directory

┌──(juhee㉿kali)-[~]
└─$ head -n 2 students.txt
Rahul
Priya

┌──(juhee㉿kali)-[~]
└─$ tail -n 2 students.txt
Riya
Rohit

┌──(juhee㉿kali)-[~]
└─$ mkdir -p backup

┌──(juhee㉿kali)-[~]
└─$ cp students.txt backup/

┌──(juhee㉿kali)-[~]
└─$ mv students.txt class_list.txt

┌──(juhee㉿kali)-[~]
└─$ grep "a" class_list.txt
Rahul
```

Priya

Aman

Riya

┌──(juhee㉿kali)-[~]

└─$ grep -n "a" class_list.txt

1: Rahul

2: Priya

3: Aman

4: Riya

┌──(juhee㉿kali)-[~]

└─$ chmod 600 class_list.txt

┌──(juhee㉿kali)-[~]

└─$ ls -l class_list.txt

-rw------- 1 juhee juhee 34 Oct 12 05:24 class_list.txt

┌──(juhee㉿kali)-[~]

└─$ find ~ -type f -name "*.txt"

/home/juhee/student.txt

/home/juhee/test/_text.txt

/home/juhee/test/vim_test.txt

/home/juhee/linux_admin_practice2/notes.txt

/home/juhee/copy.txt

/home/juhee/myprogram.txt

/home/juhee/report/notes.txt

/home/juhee/linuxlab/docs/notes.txt

/home/juhee/linuxlab/docs/report.txt

/home/juhee/linuxlab/docs/todo.txt

/home/juhee/linuxlab/backup/employees.txt

/home/juhee/linuxlab/backup/docs_copy/notes.txt

/home/juhee/linuxlab/backup/docs_copy/report.txt

/home/juhee/linuxlab/backup/docs_copy/todo.txt

/home/juhee/linuxlab/data/employees.txt

/home/juhee/myfristlab/b.txt

/home/juhee/myfristlab/a.txt

/home/juhee/backup/students.txt

/home/juhee/myfile.txt

/home/juhee/class_list.txt

/home/juhee/project/notes.txt

/home/juhee/juhee.txt

┌──(juhee㉿kali)-[~]

└─$ wc -l class_list.txt

5 class_list.txt

┌──(juhee㉿kali)-[~]

└─$ head -n 5 / etc /passwd

==> / <==

head: error reading '/': Is a directory

head: cannot open 'etc' for reading: No such file or directory

head: cannot open '/passwd' for reading: No such file or directory

```
┌──(juhee㉿kali)-[~]
└─$ tail -n 5 /var/log/dpkg.log

┌──(juhee㉿kali)-[~]
└─$ cp -r project/ project_backup/

┌──(juhee㉿kali)-[~]
└─$ less filename.txt
filename.txt: No such file or directory

┌──(juhee㉿kali)-[~]
└─$ mkdir -p archive

┌──(juhee㉿kali)-[~]
└─$ mv a.txt b.txt archive/
mv: cannot stat 'a.txt': No such file or directory
mv: cannot stat 'b.txt': No such file or directory

┌──(juhee㉿kali)-[~]
└─$ mkdir -p logs

┌──(juhee㉿kali)-[~]
└─$ for i in {1..10}; do echo "This is dummy log line $i"; done>
zsh: parse error near `\n'

┌──(juhee㉿kali)-[~]
```

└─$ for i in {1..10}; do echo "This is dummy log line $i"; done> logs/app.log

┌──(juhee㉿kali)-[~]

└─$ tail -n 3 logs/app.log

This is dummy log line 8

This is dummy log line 9

This is dummy log line 10

┌──(juhee㉿kali)-[~]

└─$ find . -type f -name "*.log" -exec cat {} +

This is dummy log line 1

This is dummy log line 2

This is dummy log line 3

This is dummy log line 4

This is dummy log line 5

This is dummy log line 6

This is dummy log line 7

This is dummy log line 8

This is dummy log line 9

This is dummy log line 10

jour2

┌──(juhee㉿kali)-[~]

└─$ grep -r "error"

.zshrc:setopt nonomatch   # hide error message if there is no match for the pattern

.zshrc: ZSH_HIGHLIGHT_STYLES[bracket-error]=fg=red,bold

grep: .cache/gstreamer-1.0/registry.x86_64.bin: binary file matches

.bashrc:# colored GCC warnings and errors

.bashrc:#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

.bashrc.original:# colored GCC warnings and errors

.bashrc.original:#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

┌──(juhee㉿kali)-[~]

└─$ touch test .sh

┌──(juhee㉿kali)-[~]

└─$ chmod +x test.sh

chmod: cannot access 'test.sh': No such file or directory

┌──(juhee㉿kali)-[~]

└─$ mkdir demo && cd demo && ls

┌──(juhee㉿kali)-[~/demo]

└─$ find ~ -type f -name "*.txt" | wc -l

23

┌──(juhee㉿kali)-[~/demo]

└─$ find /var/log -type d

/var/log

/var/log/apache2

/var/log/gvm

/var/log/apt

/var/log/inetsim

find: ‘/var/log/inetsim’: Permission denied

/var/log/notus-scanner

/var/log/redis

/var/log/mysql

/var/log/private

find: ‘/var/log/private’: Permission denied

/var/log/lightdm

find: ‘/var/log/lightdm’: Permission denied

/var/log/postgresql

/var/log/journal

/var/log/journal/2cb2453c68814152b7669f80c39c2df6

/var/log/mosquitto

find: ‘/var/log/mosquitto’: Permission denied

/var/log/nginx

/var/log/speech-dispatcher

find: ‘/var/log/speech-dispatcher’: Permission denied

/var/log/sysstat

/var/log/openvpn

/var/log/samba

/var/log/installer

/var/log/installer/cdebconf

/var/log/stunnel

/var/log/runit

/var/log/runit/ssh

/var/log/runit/pcscd

┌──(juhee㉿kali)-[~/demo]

└─$ grep -i"ext4" /etc/fstab

```
UUID=afcecf1c-e9f5-471a-b8a8-d68789d1f152 /               ext4    errors=remount-ro 0       1


┌──(juhee㉿kali)-[~/demo]
└─$ du - sh /var/log
du: cannot access '-': No such file or directory
du: cannot access 'sh': No such file or directory
4       /var/log/apache2
4       /var/log/gvm
244     /var/log/apt
du: cannot read directory '/var/log/inetsim': Permission denied
4       /var/log/inetsim
4       /var/log/notus-scanner
4       /var/log/redis
4       /var/log/mysql
du: cannot read directory '/var/log/private': Permission denied
4       /var/log/private
du: cannot read directory '/var/log/lightdm': Permission denied
4       /var/log/lightdm
4       /var/log/postgresql
40612   /var/log/journal/2cb2453c68814152b7669f80c39c2df6
40616   /var/log/journal
du: cannot read directory '/var/log/mosquitto': Permission denied
4       /var/log/mosquitto
4       /var/log/nginx
du: cannot read directory '/var/log/speech-dispatcher': Permission denied
4       /var/log/speech-dispatcher
4       /var/log/sysstat
4       /var/log/openvpn
```

```
4       /var/log/samba
14940   /var/log/installer/cdebconf
17316   /var/log/installer
4       /var/log/stunnel
4       /var/log/runit/ssh
4       /var/log/runit/pcscd
12      /var/log/runit
60024   /var/log
```

```
┌──(juhee㉿kali)-[~/demo]
└─$ ls -lS /path/to/directory | head -n 4
ls: cannot access '/path/to/directory': No such file or directory
```

```
┌──(juhee㉿kali)-[~/demo]
└─$ ped
Command 'ped' not found, did you mean:
  command 'pwd' from deb coreutils
  command 'red' from deb ed
  command 'ed' from deb ed
  command 'med' from deb ncl-ncarg
  command 'pev' from deb plc-utils-extra
  command 'pem' from deb pem
  command 'psd' from deb profile-sync-daemon
  command 'jed' from deb jed
  command 'pee' from deb moreutils
  command 'pdd' from deb pdd
  command 'peg' from deb peg
```

```
  command 'zed' from deb zfs-zed
  command 'bed' from deb bed
  command 'pen' from deb pen
  command 'sed' from deb sed
Try: sudo apt install <deb name>

┌──(juhee㉿kali)-[~/demo]
└─$ pwd
/home/juhee/demo

┌──(juhee㉿kali)-[~/demo]
└─$ ls -la
total 8
drwxrwxr-x  2 juhee juhee 4096 Oct 12 06:16 .
drwx------ 29 juhee juhee 4096 Oct 12 06:16 ..

┌──(juhee㉿kali)-[~/demo]
└─$ touch filename.txt

┌──(juhee㉿kali)-[~/demo]
└─$ printf "Apple\nBanana\nApple\nMango\nBanana\n" > fruits.txt sort fruits.txt | uniq
Apple
Banana
Apple
Mango
Banana
```

┌──(juhee㉿kali)-[~/demo]

└─$ uname -a > sys_info.txt

┌──(juhee㉿kali)-[~/demo]

└─$ cat sys_info.txt

Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux

┌──(juhee㉿kali)-[~/demo]

└─$ find /var/log -type f -mtime -7

/var/log/Xorg.0.log.old

/var/log/boot.log.1

find: ‘/var/log/inetsim’: Permission denied

/var/log/vmware-vmusr-juhee.log

/var/log/vmware-network.log

/var/log/Xorg.0.log

/var/log/boot.log

/var/log/vmware-vmsvc-root.1.log

/var/log/vmware-vmtoolsd-juhee.log

find: ‘/var/log/private’: Permission denied

find: ‘/var/log/lightdm’: Permission denied

/var/log/journal/2cb2453c68814152b7669f80c39c2df6/user-1000.journal

/var/log/journal/2cb2453c68814152b7669f80c39c2df6/system.journal

find: ‘/var/log/mosquitto’: Permission denied

/var/log/vmware-vmsvc-root.log

/var/log/wtmp

find: ‘/var/log/speech-dispatcher’: Permission denied

/var/log/vmware-vmtoolsd-root.log

/var/log/macchanger.log.1.gz

┌──(juhee㉿kali)-[~/demo]

└─$